CODE-LOGIK
Crafting Cutting-Edge Software

THE WILLOW TREE OMS

CS4233 CAPSTONE PROJECT 2024

PROJECT PROPOSAL

PREPARED BY:

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PROJECT OVERVIEW

PROJECT NAME: The Willow Tree OMS
PROJECT REFERENCE: CS4233 CAPSTONE PROJECT 2024
DATE: 23 JAN 2024

The Willow Tree's current Order Management System (OMS) allows customers to place orders at the table via a kiosk.

Although the current release is functional, the OMS limits menu items to those available during development. A redesign is necessary because the menu now features previously unavailable items.

The current OMS design has embedded all menu items in the source code and requires a redesign to accommodate future menu items.

Code-Logik proposes translating the OMS code to a more efficient language to accommodate highly interactive and dynamic content, migrating the menu items to a database for menu customization, integrating logic to connect to the database, and creating a privileged on-demand user interface to manage menu items.

The proposal aims to deliver The Willow Tree an easily accessible and manageable OMS capable of keeping pace with evolving menu items and providing the customer with the best possible dining experience.

To complete the redesign of The Willow Tree OMS, Code-Logik estimates a total of 13 weeks from start to finish.

SCOPE OF WORK

With The Willow Tree's approval, Code-Logik will perform the following scope of work to redesign the current OMS:

- Translate the current Front-End code to an efficient language capable of handling the highly interactive nature of the OMS in a restaurant setting.

- Migrate the current menu items embedded in the source code into a database to create a customizable, fully dynamic OMS.

- Integrate the Back-End logic necessary to connect with a database.

- Create a privileged Front-End user interface to allow updates to the menu without the need for a full-scale redesign, as is the current case.

TIMELINE

The scope of work will follow a five-phase structured process typical of the Software Development Life Cycle (SDLC). Figure 1 details the expected timeline for each phase of the SDLC.

Phases	Date of Phases (by Weeks)												
1. Requirements Analysis													
2. Modeling													
3. Testing													
4. Source Code													
5. Published on GitHub													
	02	09	16	23	01	08	15	22	29	05	12	19	29
	February				March					April			

Figure 1 Expected Timeline for Software Development Life Cycle.

Phase Objectives:

1. **Requirements Analysis:** gathering, evaluating, and documenting the requirements to redesign the OMS will take approximately two weeks.

2. **Modeling:** systematically organizing abstractions of key OMS components for decision-making purposes will take approximately four weeks.

3. **Testing:** verifying key OMS components and validating redesign requirements will take approximately eight weeks.

4. **Source Code:** creating standardized, understandable, and documented source code will take approximately eight weeks.

5. **Published on GitHub:** Delivering requirements met redesigned OMS software via GitHub will take approximately two weeks.

BUDGET

The estimated budget for this project is $25,000. The budget includes all requirements analysis, modeling, testing, source code, and publishing costs. Upon taking ownership, all the expenses related to support services will be the sole responsibility of The Willow Tree.

TEAM MEMBERS

Mark Sarasua, Jr. will be the sole developer on this project and will fulfill all roles and responsibilities.

RISK MANAGEMENT PLAN

Risk identification and assessment will occur during phase 1 of the SDLC, and a risk register will document the results. Regular updates to the risk register will occur throughout the SDLC. The risk register will include a mitigation plan for each risk listed.

QUALITY ASSURANCE PLAN

To ensure the redesigned OMS meets the requirements and expectations of The Willow Tree and its customers, throughout the SDLC, use of various quality assurance and quality control activities such as code analysis, unit testing, integration testing, system testing, user acceptance testing, and issue tracking will take place during all phases where appropriate.

ACCEPTANCE CRITERIA

Functional Requirements: The redesigned OMS user interface should be able to adjust dynamically according to the number of active menu items in the database.

Performance Requirements: The redesigned OMS should consistently function properly regardless of the number of menu item updates.

Usability Requirements: The redesigned OMS should be easy to use with an aesthetically pleasing user interface.

Security Requirements: The redesigned OMS should be secure and protect the menu items database from unauthorized access.

Compatibility Requirements: The redesigned OMS should be compatible with industry-standard operating system environments.

Maintenance Requirements: The redesigned OMS should be easy to support and update.

CONCLUSION

Redesigning the Order Management System into a customizable dynamic OMS is the goal of this project. The OMS can reliably serve the embedded menu items. However, the OMS provides no flexibility in adding new menu items. Code-Logik will redesign the OMS into a manageable application without requiring a programming expert to customize the menu items. This project will undoubtedly provide The Willow Tree's customers with the best possible dining experience by allowing It to adjust its menu anytime.

Once the proposal is approved, Code-Logik will start "Crafting Cutting-Edge Software" for The Willow Tree.

For further details and information, please contact:

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